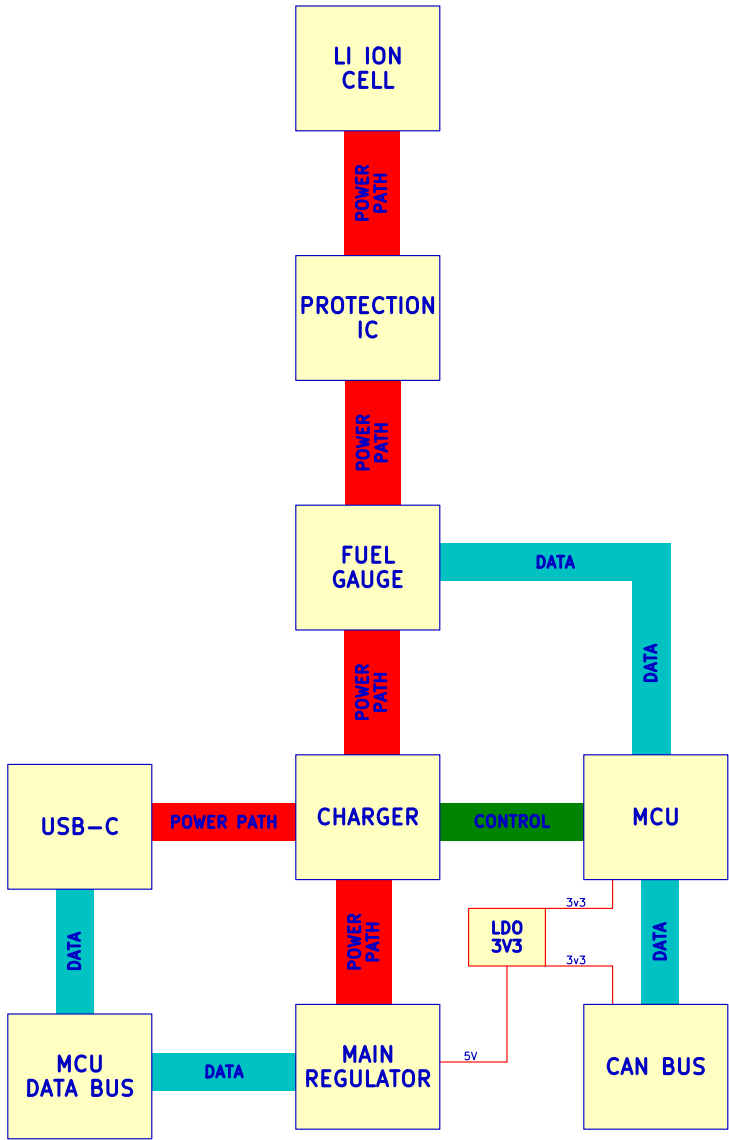
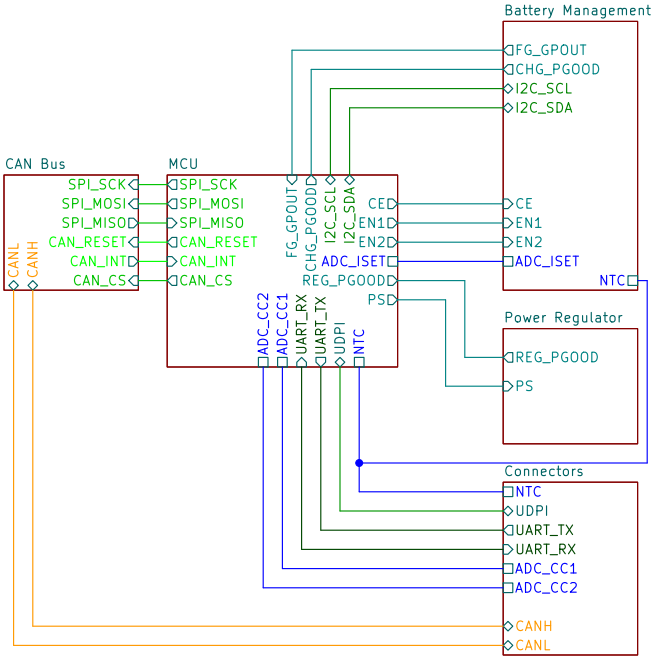


5W 2700 mAh Power Supply with USB Charging
w/ Optional Fuel Gauge & CAN Bus



github.com/FETfox/Li-Ion_Power_Module

Sheet: /

File: power_module.kicad_sch

Title: Li Lion Power Module w/ Optional Fuel Gauge & CAN Bus

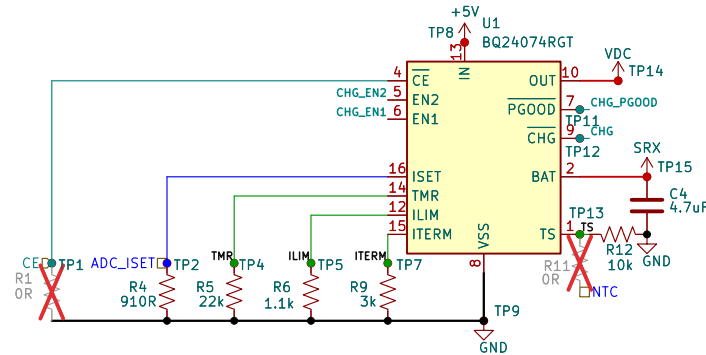
Size: USLetter Date: 2026-06-25

Rev: RevA

KiCad E.D.A. 10.0.5

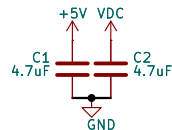
Id: 1/6

Li Ion Cell, Charger IC, Protection IC & Optional Fuel Gauge IC

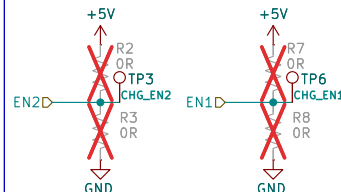


BQ24074 Charge Controller w/ Power Path

Place decoupling capacitors as close to power pins as possible.

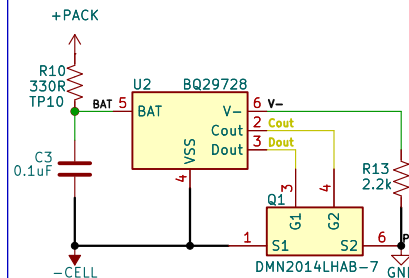


BQ24074 Decoupling Capacitors



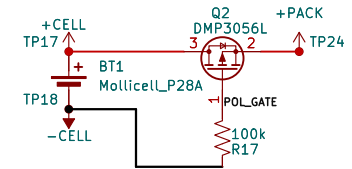
EN pin settings

EN2	EN1	INPUT CURRENT
0	0	100mA
0	1	500mA
1	0	Set by ILIM
1	1	Standby



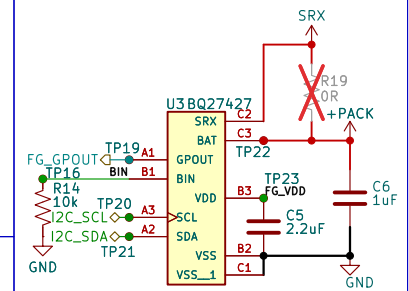
BQ29728 Cell Protection Circuit

Then is PMOS being used for polarity protection.

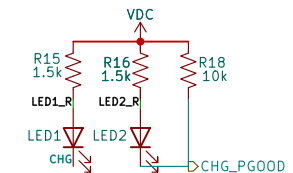


Li Ion Cell

R19 serves as a jumper for when the BQ27427 Fuel Gauge is not populated.



BQ27427 Fuel Gauge



BQ24074 LED Pullups

github.com/FETfox/Li-Ion_Power_Module

Sheet: /Battery Management/
File: batt_mgmt.kicad_sch

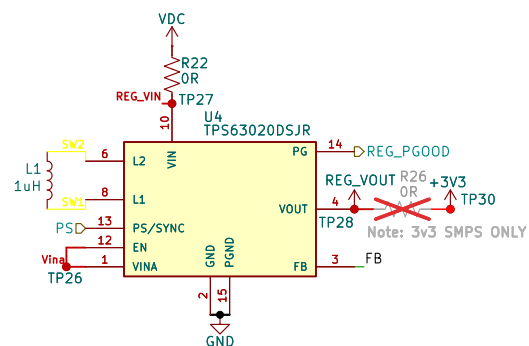
Title: Li Lion Power Module w/ Optional Fuel Gauge & CAN Bus

Size: USLetter Date: 2026-06-25
KiCad E.D.A. 10.0.5

Rev: RevA
Id: 2/6

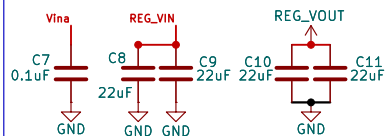
Boostbuck Regulator & Optional 3v3 LDO

For 5v output, 3v3 LDO must be placed in series via populating R34.



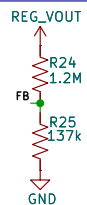
TPS63020 Power Supply Regulator

Place decoupling capacitors physically as close as possible to their corresponding pin.

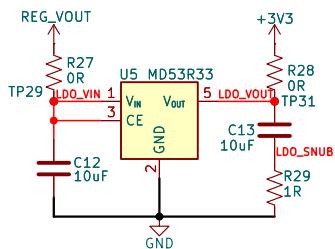


TPS63020 Decoupling Capacitors

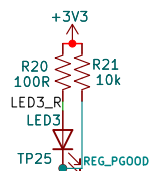
Voltage output is set by top resistor:
768k resistor for 3v3 output
1.2M resistor for 5v



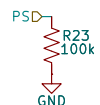
TPS63020 GND
Voltage feedback divider



MD53R33 3v3 LDO Regulator

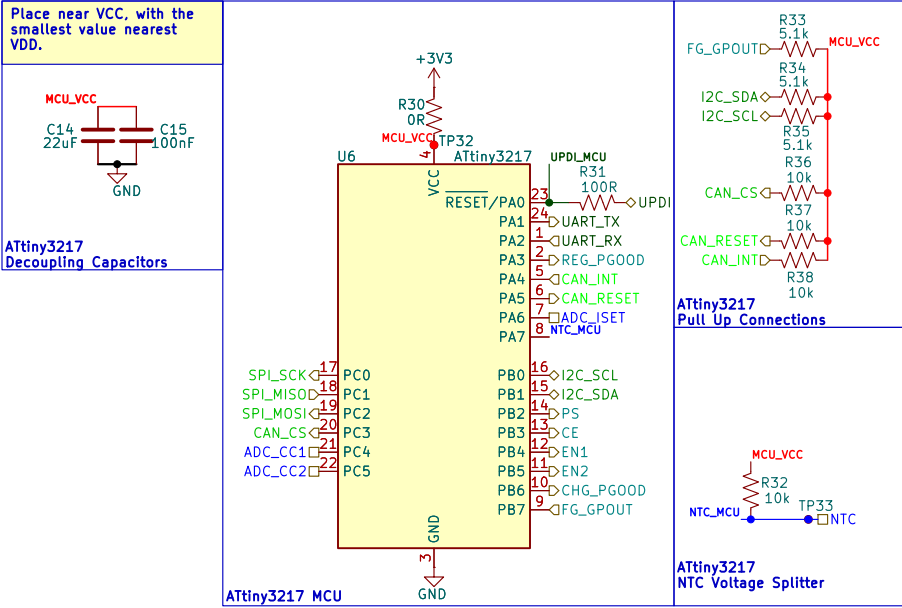


TPS63020 PGOOD Pullup



TPS63020 PS Pulldown

MCU



github.com/FETfox/Li-Ion_Power_Module

Sheet: /MCU/

File: mcu.kicad_sch

Title: Li Lion Power Module w/ Optional Fuel Gauge & CAN Bus

Size: USLetter

Date: 2026-06-25

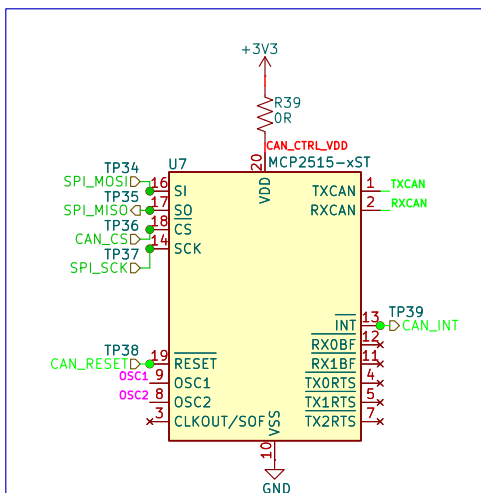
Rev: RevA

KiCad E.D.A. 10.0.5

Id: 4/6

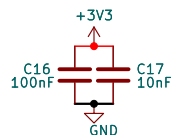
CAN Bus Interface

CAN Controller



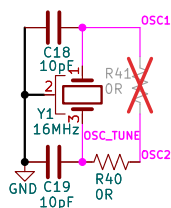
MCP2515 CAN Controller

Place near VDD, with the smallest value nearest VDD.



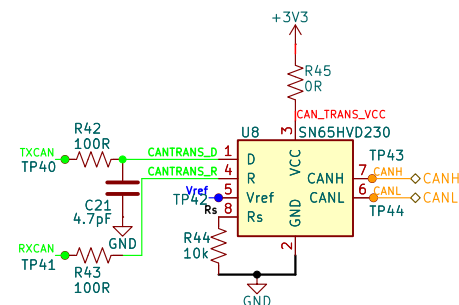
MCP2515 Decoupling Capacitors

Place as close to OSC pins as possible.



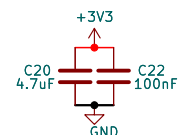
MCP2515 Oscillator Circuit

CAN Transceiver



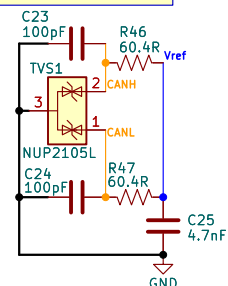
SN65HVD230 CAN Transceiver

Place near VCC, with the smallest value nearest VCC.



SN65HVD230 Decoupling Capacitors

Place as close to
SN65HVD230 as possible.



SN65HVD230 EMI Filtering & Termination

Connectors

